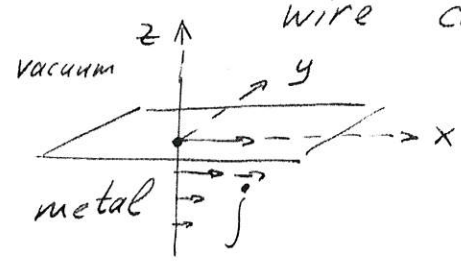
Problem 1 when $\delta \ll a$ the surface of the wire can be approximated as flat



$\vec{j} = \sigma_0 \vec{E}$

both $\vec{j}$ and $\vec{E}$ are along x.

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = i\omega \vec{B}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j} = \mu_0 \sigma_0 \vec{E}$$ (displacement current is neglected as a small correction at $\omega \ll \omega_{pl}$)

$$\nabla \times (\nabla \times \vec{E}) = i\omega \mu_0 \sigma_0 \vec{E}$$

Since $\nabla \cdot \vec{E} = 0$ (why?) $\quad \nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$

$$\nabla^2 \vec{E} = -i\omega \mu_0 \sigma_0 \vec{E}$$

In our setup $\vec{E} = (E(z), 0, 0)$

$$\frac{d^2 E}{dz^2} = -i\omega \mu_0 \sigma_0 E$$ The solution, decaying when $z \to -\infty$ is

$$E(z) = E_0 e^{\sqrt{-i\omega\mu_0\sigma_0}\, z} = E_0 e^{\frac{z}{\delta}} e^{-\frac{iz}{\delta}}, \quad \delta = \sqrt{\frac{2}{\omega \mu_0 \sigma_0}}$$

as $\sqrt{-i} = \frac{-1-i}{\sqrt{2}}, \frac{1-i}{\sqrt{2}} \Rightarrow$ the second value should be chosen

We can now write actual electric field as the real part of the above solution:

$$E(z,t) = E_0 \, \mathrm{Re}\, e^{z/\delta} e^{-iz/\delta} e^{-i\omega t} =$$

$$= \rho_0 j_0 e^{z/\delta} \cos(\omega t + z/\delta), \quad \text{where } \rho_0 = \frac{1}{\sigma_0}$$

we introduced the amplitude of current j_0 at

the surface. It can be related to the total current via

$$I_0 = 2\pi a j_0 \int_{-\infty}^{0} dz\, e^{z/\delta} \cos\left(\frac{z}{\delta}\right) =$$

$$= 2\pi a j_0 \, \mathrm{Re} \int_{-\infty}^{0} dz\, e^{\frac{z}{\delta}(1-i)} =$$

$$= 2\pi a j_0 \, \mathrm{Re} \left(\frac{\delta}{1-i}\right) = \pi \delta a \quad \Rightarrow \boxed{j_0 = \frac{I_0}{\pi a \delta}}$$

Thus,

$$E(z,t) = \rho_0 \frac{I_0}{\pi a \delta} e^{z/\delta} \cos(\omega t + z/\delta)$$

Total Joule heat: per unit length:

$$Q = 2\pi a \int_{-\infty}^{0} dz\, \overline{E(z,t) \cdot j(z,t)} = \frac{2\pi a}{\rho_0} \int_{-\infty}^{0} dz \overline{\left(E(z,t)\right)^2}$$

$$= \frac{2\pi a}{\rho_0} \left(\frac{\rho_0 I_0}{\delta \pi a}\right)^2 \int_{-\infty}^{0} dz\, e^{2z/\delta} \overline{\cos^2(\omega t + z/\delta)}$$

The time average is simply $\overline{\cos^2(\omega t + z/\delta)} = \frac{1}{2}$

and $\int_{-\infty}^{0} dz\, e^{2z/\delta} = \frac{\delta}{2}$, so that

$$\boxed{Q = \frac{\rho_0 I_0^2}{\pi a \delta} = \frac{I_0^2}{\pi a \delta \sigma_0}}$$

Problem 2

$$\vec{j} = \sigma(\omega)\vec{E} = \frac{\sigma_0}{1-i\omega\tau}\vec{E} \qquad \text{(Ohm's law)}$$

$$\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \qquad \text{(Gauss' law)}$$

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{j} = 0 \Rightarrow i\omega\rho = \nabla\cdot\vec{j} \quad \text{(continuity of charge flow)}$$

Excluding ρ with the help of the last equation $\rho = -\frac{i\nabla\cdot j}{\omega}$ and using it in the second one:

$$\nabla\cdot\vec{E} = -\frac{i}{\varepsilon_0\omega}\nabla\cdot\vec{j} \qquad (*)$$

On the other hand taking $\nabla\cdot$ operation to both sides of the Ohm's law we get

$$\nabla\cdot\vec{j} = -\frac{\sigma_0}{1-i\omega\tau}\nabla\cdot\vec{E} \qquad (**)$$

These two last equations are compatible only if

$$\frac{\sigma_0}{1-i\omega\tau}\times\frac{i}{\omega\varepsilon_0} = -1$$

This is a quadratic equation:

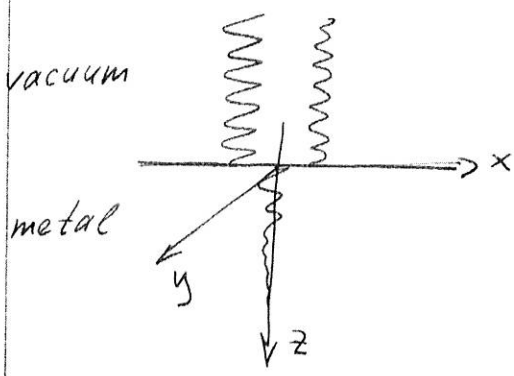
$$\omega\varepsilon_0 - i\omega^2\tau\varepsilon_0 = i\sigma_0 = -i\varepsilon_0\omega_p^2\tau$$

or $\boxed{\omega^2 = \omega_p^2 - \frac{i\omega}{\tau}}$ In the limit of $\omega_p\tau \gg 1$ the last term is small in which ω can be replaced with ω_p to the lowest order:

$$\omega^2 = \omega_p^2 - \frac{i\omega_p}{\tau} \Rightarrow \omega = \sqrt{\omega_p^2 - \frac{i\omega_p}{\tau}} \approx \omega_p - \frac{i}{2\tau}$$

$$\omega' = \omega_p \quad \text{and} \quad \omega'' = -1/2\tau \ll \omega'$$

Problem 3



Incident + reflected wave

$$B_y(z<0) = B_0 e^{-i\omega t + \frac{i\omega}{c} z} + B_r e^{-i\omega t - \frac{i\omega}{c} z} \qquad (1)$$

Transmitted (evanescent) wave

$$B_y(z>0) = B_t e^{-i\omega t} e^{i\frac{\omega}{c} n z} \qquad (2)$$

where $n = n' + i n'' = (1+i)\sqrt{\frac{\sigma}{2\varepsilon_0 \omega}}$ (see the notes)

Note that when σ is purely real, $n' = n''$.

To find dissipation we first need to find electric field $\vec{E} = (E_x, 0, 0)$ inside the metal

$$(\nabla \times \vec{E})_y = i\omega B_y$$

$$\frac{\partial E_x}{\partial z} = i\omega B_y$$

which together with Eq. (2) gives

$$E_x(z>0) = \frac{c B_t}{n} e^{-i\omega t + \frac{i\omega}{c} n z}$$

From the notes $B_t = \frac{2n B_0}{1+n}$ (B_0 is real)

Taking the real part

$$E_x = 2cB_0 \, \mathrm{Re} \, \frac{e^{-i\omega t + \frac{i\omega n}{c} z}}{1+n} =$$

$$= 2cB_0 \, \mathrm{Re} \, \frac{(1+n'-in'')e^{-i\omega t + \frac{i\omega n}{c} z}}{(1+n')^2 + n''^2}$$

$$E_x = 2cB_0 \frac{(1+n')\cos(\omega t - \frac{n'\omega}{c}z) - n''\sin(\omega t - \frac{n'\omega}{c}z)}{(1+n')^2 + n''^2} e^{-\frac{\omega n''}{c}z}$$

Joule heat (per unit area) is

$$Q = Z_0 \int_0^\infty dz \, \overline{E_x^2(z,t)}$$

Averaging over time is performed using

$$\overline{\cos^2\omega t} = \overline{\sin^2\omega t} = \frac{1}{2}$$

$$\overline{\cos\omega t \cdot \sin\omega t} = 0$$

and yields

$$\boxed{Q = 4c^2B_0^2 Z_0 \frac{c}{2\omega n''} \frac{1}{(1+n')^2 + n''^2}} \qquad (3)$$

The incident energy flux is

$$S = \frac{E_0 B_0}{\mu_0} = \boxed{\frac{c B_0^2}{\mu_0}} \qquad (4)$$

So far our discussion leading to Eq. (3) has been completely general. Let us now take into account that in our model $n' = n''$ and that $Z_0 = 2\varepsilon_0 \omega n''^2$

Eq. (3) now gives

$$Q = 4c^2B_0^2 \frac{\varepsilon_0 n'' c}{(1+n'')^2 + n''^2} = \underbrace{\frac{cB_0^2}{\mu_0}}_{S} \frac{\overset{1}{4c^2\mu_0\varepsilon_0} n''}{(1+n'')^2 + n''^2}$$

or

$$\boxed{\frac{Q}{S} = \frac{4n''}{1+2n''+2n''^2}} \checkmark$$

Comparing now this with R from the notes (when $n' = n''$):

$$1 - R = 1 - \frac{(n'-1)^2 + n''^2}{(n'+1)^2 + n''^2} = \frac{4n'}{(1+n')^2 + n''^2}$$

$$= \boxed{\frac{4n''}{1+2n''+2n''^2} \equiv \frac{Q}{S}}$$